Xinze (Jason) Chen

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Education

PhD	**Pennsylvania State University**, Biomedical Sciences	Aug 2026 – Jun 2031
MS	**University of California, San Diego**, Data Science	Sep 2023 – Jun 2026
BS	**University of California, San Diego**, Neurobiology & Cognitive Science with a Specialization in Machine Learning and Neural Computation	Sep 2019 – Jun 2023

Publications

2026

Green, A.[*], Vashist, T.[*], Jakhmola, S., **Chen, X.**[†], Baidwan, G.[†], Buchanan, J.[†], Tiwari, S., Griffin, E., Howell, A., Lee, Y., Moore, D., Gianella, S., Smith, D., Zhu, Q., Walss-Bass, C., Wang, A.[‡], Mukamel, E. A.[‡], Gaulton, K. J.[‡], Rana, T. M.[‡]. ***Cell-type specific impact of opioid use disorder and HIV on the human forebrain and cerebellum***.

- Manuscript under review at *Nature*.
- [*]Co-first authors; [†]Co-second authors; [‡]Co-corresponding authors.

Research Experience

University of California, San Diego, Graduate Student Researcher — La Jolla, CA, Jan 2024 – Present

- Principal investigator: Eran Mukamel, PhD
- Conduct research as part of the SCORCH (Single Cell Opioid Responses in the Context of HIV) consortium investigating the epigenomic and transcriptomic effects of HIV and opioid use disorder on the human brain
- Build standardized pipeline to process and analyze multiomic sequencing data
- Analyze single-nucleus RNA sequencing (snRNA-seq) and single-nucleus ATAC sequencing (snATAC-seq) datasets to study gene expression and chromatin accessibility in brain regions such as the amygdala, prefrontal cortex, and cerebellum
- Identify differentially expressed genes and chromatin accessibility regions, uncovering genetic and epigenetic markers relevant to neurological diseases
- Integrate chromatin accessibility and gene expression data for inferring transcription factor regulatory networks
- Identify co-regulated gene modules, and cell-type-specific transcriptional programs disrupted by HIV and opioid exposure

University of California, San Diego, Graduate Student Researcher — La Jolla, CA, Jul 2025 – Sep 2025

- Principal investigator: Benjamin Smarr, PhD
- Process and analyze continuous glucose monitoring (CGM) data

Scripps Research, Research Intern — La Jolla, CA, Sep 2022 – Jun 2023

- Principal investigator: Lisa Stowers, PhD
- Applied machine learning models to analyze neural activity and facial movements
- Utilized machine learning algorithms and traditional data science techniques to process and analyze time-series data from mouse experiments
- Produced valuable insights into mice neural circuitry and facial expression associated with social behaviors

Research Project

Revealing the single cell determinants of brain relevant to persistent HIV infection and opioid use disorder — Feb 2024 - Present

UCSD SCORCH Rana, 1U01DA053630-01

- Principal investigators: Tariq Rana, PhD & Eran Mukamel, PhD & Kyle Gaulton, PhD & Allen Wang, PhD
- Conduct research as part of the UCSD SCORCH investigating the impact of HIV and opioid use disorder (OUD) on the brain
- Utilize single-nucleus RNA sequencing (snRNA-seq) and single-nucleus ATAC sequencing (snATAC-seq) to analyze gene expression and chromatin accessibility
- Perform integrative gene regulatory network (GRN) analysis to identify transcriptional rewiring across HIV and OUD conditions
- Aim to uncover molecular mechanisms underlying neuropsychiatric disorders and identify potential therapeutic targets

Posters & Presentations

HIV and Opioid Use Disorder: A Single-Cell Multiomic Study of the Human Amygdala — Jun 2024

Chen, X., Green, A., Buchanan, J., Lee, Y., Mummey, H., Kudtarkar, P., et al.

- Poster presented at the Summer SCORCH Consortium Meeting, Baltimore, MD, June 2024
- Conducted single-cell multiomic analysis of transcriptomic and epigenomic changes in the amygdala related to HIV and OUD
- Identified key cell types and differentially expressed genes, providing insights into molecular mechanisms behind neuropsychiatric symptoms

The Transcriptomic and Epigenomic Impacts of HIV and Opioid Use Disorder Across Human Prefrontal Cortex Cell Types — Jun 2024

Green, A., **Chen, X.**, Buchanan, J., Mummey, H., Lee, Y., Kudtarkar, P., et al.

- Poster presented at the Summer SCORCH Consortium Meeting, Baltimore, MD, June 2024
- Investigated effects of HIV and OUD on gene expression and chromatin accessibility in the prefrontal cortex (PFC)
- Highlighted 135 differentially expressed genes with immune and transcriptional regulatory changes in oligodendrocytes and microglia

Teaching Experience

University of California, San Diego, Teaching Assistant — La Jolla, CA

COGS 118A: Supervised Machine Learning Algorithms & COGS 118B: Introduction to Machine Learning — Jul 2024 – Sep 2024, Jul 2025 - Sep 2025

- Ensured smooth course operations, provided timely student support, and streamlined grading processes
- Developed scripts to manage late penalties automatically and sync grades from GradeScope to Canvas

University of California, San Diego, Teaching/Instructional Assistant — La Jolla, CA

COGS 17: Neurobiology of Cognition — Sep 2021 - Dec 2021, Jan 2024 – Mar 2024, Mar 2024 – Jun 2024, Sep 2024 – Dec 2024,

- Assisted with course materials, and prepared personalized materials for discussion sections
- Held office hours to clarify difficult concepts, facilitated group discussions, and

helped organize learning materials

- Improved student engagement and comprehension

Jan 2025 - Mar 2025
Jan 2026 - Mar 2026

University of California, San Diego, Teaching/Instructional Assistant
COGS 108: Data Science in Practice

La Jolla, CA
Sep 2022 – Dec 2022,
Sep 2023 – Dec 2023,
Mar 2025 - Jun 2025,
Sep 2025 - Dec 2025

- Assisted with course materials and automated grading processes to improve efficiency
- Developed and optimized auto-grading scripts and provided hands-on support during office hours and discussion sections

Technical Skills

Data Science, Bioinformatics, & Machine Learning: Python, R, MATLAB, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, ScanPy, Seurat, ArchR, SnapATAC2, SCENIC+, chromVAR, MACS3, bedtools, samtools, deepTools

Single-Cell & Genomics Analysis: snRNA-seq preprocessing/QC, snATAC-seq peak calling and integration, multiomic (RNA–ATAC) integration, transcription factor motif analysis, gene regulatory network (GRN) inference, differential expression and accessibility analysis, BigWig generation, genome browser visualization

Computational & Statistical Skills: High-dimensional data analysis, nonlinear dimensionality reduction, Leiden/Louvain clustering, signal processing, time-series analysis, mixed models and GLMs, data visualization (matplotlib, seaborn, ggplot2)

Software & Tools: Git, Jupyter, LaTeX, SLURM/HPC environments, Docker, VS Code